Floating point numbers: Rounding Issues

CSE 101 Section 03

Introduction to Algorithms and Computing

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Introduction

As discussed in the previous lecture, the mantissa of a floating-point numbers can only be stored in a limited number of bits.

e.g. for single precision:

if more than 23 bits are needed for a given number, precision is lost…

Rounding errors examples

```
x=1/6;
fprintf("x= %f\n",x);
fprintf("Check1\n");
if 1 == x+x+x+x+x+x
        fprintf("Check1 is True\n");
else
        fprintf("Check1 is False with difference of %e\n",1-(x+x+x+x+x+x));
end
```

Rounding errors examples

```
time = 0;
for k = 1:100000
        time = time + 0.1;
end


if time == 10000
        disp('Time to launch the missile.')
end
```

What is a rounding error?

- Rounding error is the difference between an approximation of a number used in computation and its exact (correct) value.

- In certain types of computation, a rounding error can be magnified as any initial errors are carried through one or more intermediate steps.

Real world examples of rounding errors

The Patriot missile defense system

- Was used during the Gulf War and was also rendered ineffective due to rounding error.
- The system used an integer timing register which was incremented at intervals of `0.1`s.
- The binary approximation of `0.1` was used

$$0.0001100110011001100110011_2 = 209715/2097152$$

Real world examples of rounding errors

The Patriot missile defense system

After 100 hours (3.6 x 106 ticks*), an error of

$$\left(\frac{1}{10} - \frac{209715}{2097152}\right)(3.6 \times 10^6) = \frac{5625}{16834} \approx 0.3433\ seconds$$

had accumulated. This discrepancy caused the Patriot system to continuously recycle itself instead of targeting properly. As a result, an Iraqi Scud missile could not be targeted and was allowed to detonate on a barracks, killing 28 people.

Real world examples of rounding errors

Ariane rocket

- launched on June 4, 1996
- In the 37th second of flight, the inertial reference system attempted to convert a 64-bit floating-point number to a 16-bit number.
  - Triggered an overflow error.
  - The error was interpreted by the guidance system as flight data.
  - Causing the rocket to veer off course and be destroyed.

Next…

Von Neumann architecture